Real Batteries and Voltmeters

Assignment 4

SCIE 001 Physics

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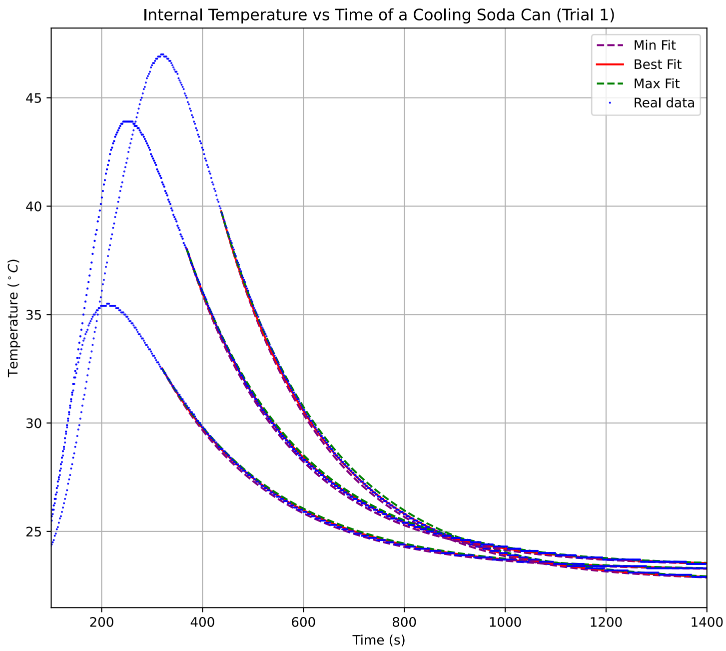

Figure 1: Graph of the internal temperature over time for Trial 4.

Contents

Appendix

Introduction and Description

> For this assignment, you essentially only need to finish what you started in tutorial, where you investigated the internal resistance of a battery and of your voltmeter. Your submission should include two figures along with a brief description of your method and discussion of your results.

1. Battery internal resistance

1.1. Methodology

> [100 words max] Briefly explain your methodology (e.g. how you chose to vary the resistance: potentiometer? resistors?, over what resistance range? how did you measure the resistance value?, how did you determine your uncertainties?)

Resistance over a 9 volt battery was varied from 1 Ohm to 2 M Ohms by combining various small resistors on a breadboard. Voltage across the resistance was then measured with a digital multimeter (DMM). To gain further accuracy and account for the resistance of the breadboard, resistances were also measured with a DMM.

Uncertainties were propagated from the uncertainty reported in the DMM's manual.

Include a reference from the manual

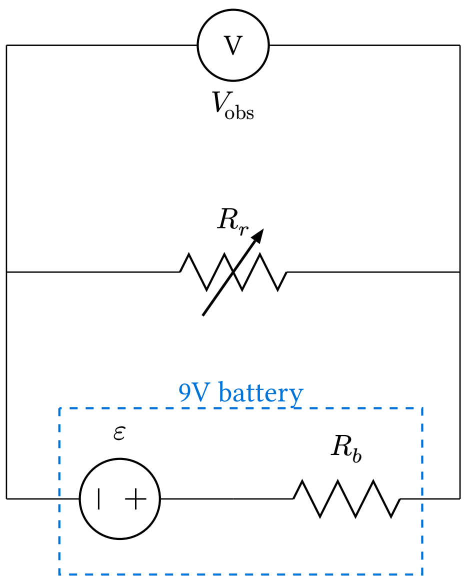

Figure 2: A circuit diagram of the setup.

"Define the variables here"

subtract these words from the count

$$V = IR$$

$$I = \frac{V_{\text{obs}}}{R_r}$$

$$u[I] = \sqrt{\left(u[V_{\text{obs}}]\left(\frac{\partial}{\partial V_{\text{obs}}}\frac{V_{\text{obs}}}{R_r}\right)\right)^2 + \left(u[R_r]\left(\frac{\partial}{\partial R_r}\frac{V_{\text{obs}}}{R_r}\right)\right)^2}$$

$$u[I] = \sqrt{\left(\frac{u[V_{\text{obs}}]}{R_r}\right)^2 + \left(u[R_r]\frac{-V_{\text{obs}}}{R_r^2}\right)^2}$$

From this:

$$V = IR$$

$$\varepsilon = I(R_r + R_b) \qquad V_{\text{obs}} = IR_r$$

$$\varepsilon - IR_r = IR_b$$

$$\varepsilon - V_{\text{obs}} = IR_b$$

$$\therefore I = -\left(\frac{1}{R_b}\right)V_{\text{obs}} + \frac{\varepsilon}{R_b}$$

Or in other words, in the linear relationship between I and V:

$$\text{Slope} = m = -\frac{1}{R_b}$$

$$\therefore R_b = -\frac{1}{m}$$

$$u[R_b] = \sqrt{\left(u[m]\left(\frac{\partial}{\partial m}\left(-\frac{1}{m}\right)\right)\right)^2}$$

$$\therefore u[R_b] = u[m]\left(\frac{1}{m^2}\right)$$

Words: 83

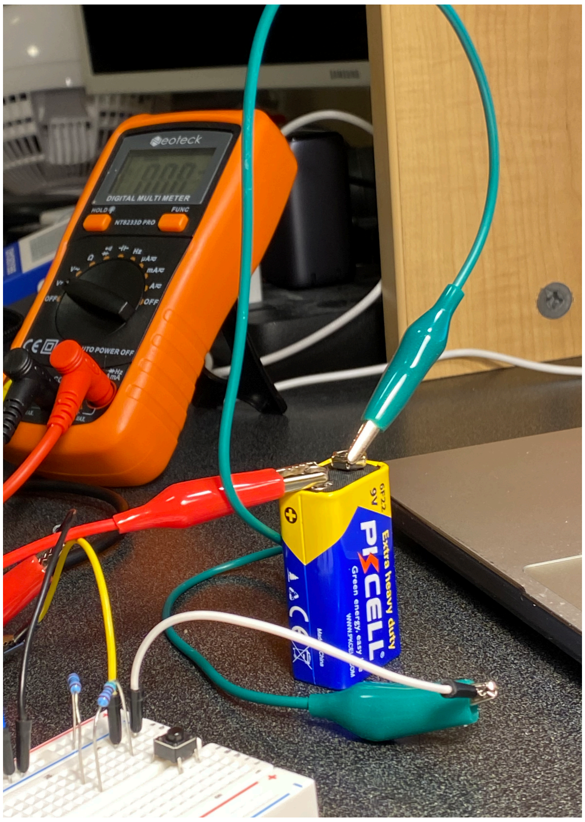

Figure 3: Image of the experimental setup.

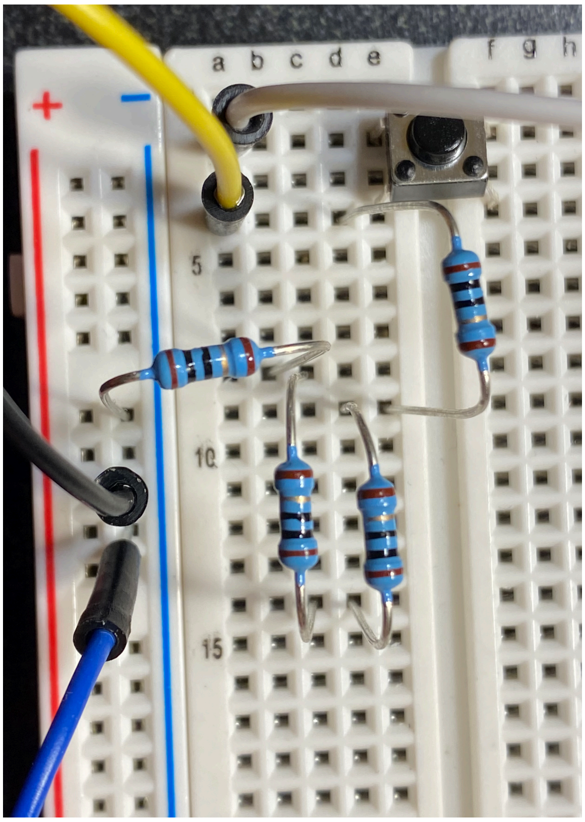

Figure 4: An image of the tissue box that was dropped onto the sensor

Fix captions

1.2. I-V Curve

> Include a figure that shows the I-V curve for a resistance of varying value connected to a 9V battery. Fit the curve and give the internal resistance of the battery in the caption. You can include fit parameters in the plot or in the caption.

1.3. Internal Resistance of Voltmeter

> [100 words max] Briefly comment on any marked deviation from the expected linear fit and if the obtained value of the internal resistance is reasonable.

Words: 0

2. Voltmeter internal resistance

2.1. Methodology

> [100 words max] Briefly explain your methodology (e.g. why you picked the capacitor you picked, how you recorded the time-dependent data, how you determined your uncertainties)

2.2. Discharge Voltage vs Time Curve

> Include a figure that shows the discharge voltage vs time of a capacitor connected to your voltmeter. Fit the curve and give the time constant and value of the internal resistance of your voltmeter in the caption. You can include fit parameters in the plot or in the caption.

2.3. Voltmeter Internal Resistance

> [100 words max] Briefly comment on if the obtained value of the internal resistance is reasonable and on the implication of your results when making measurements with your voltmeter.

Words: 0

A Interactive Fitter

Note: I included this app